

## The logic of arguments - Set 2

TMUA practice questions by topic. Specification Arg1-Arg4. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Consider the statement  $S$ : **for all real  $x$ ,  $x^2 - 6x + 10 > 0$** . Which of the following is true?
- A  $S$  is false, because  $x = 3$  gives  $x^2 - 6x + 10 = 0$ .
  - B  $S$  cannot be decided without knowing  $x$ .
  - C  $S$  is false, because a quadratic can be negative.
  - D  $S$  is true, because  $x^2 - 6x + 10 = (x - 3)^2 + 1$ .
  - E  $S$  is true, because  $x^2 - 6x + 10$  has two real roots.
- 2 For real numbers  $a, b$ :  $P$ : ' $ab = 0$ ';  $Q$ : ' $a = 0$ '. Which one of the following is true?
- A  $P$  is necessary and sufficient for  $Q$ .
  - B  $P$  is necessary but not sufficient for  $Q$ .
  - C  $P$  is sufficient but not necessary for  $Q$ .
  - D  $P$  is neither necessary nor sufficient for  $Q$ .
- 3 Consider the statement: **if  $a$  and  $b$  are irrational, then  $a + b$  is irrational**. Which of the following pairs is a counterexample?
- A  $a = 1, b = -1$
  - B  $a = \sqrt{2}, b = -\sqrt{2}$
  - C  $a = \pi, b = \sqrt{2}$
  - D  $a = \sqrt{2}, b = 2\sqrt{2}$
  - E  $a = \sqrt{2}, b = \sqrt{3}$
- 4 Let  $S$  be the statement: **every even number greater than 2 can be written as the sum of two primes**. A single counterexample to  $S$  would be
- A an odd number that is not the sum of two primes
  - B an even number greater than 2 that can be written as the sum of two primes in two different ways
  - C two primes whose sum is odd
  - D an even number that is the sum of two primes
  - E an even number greater than 2 that is not the sum of any two primes

- 5 For a real number  $x$ ,  $P$ : ' $x > 0$ '.  $Q$ : ' $x^3 > 0$ '. Which one of the following is true?
- A P is necessary and sufficient for Q.
  - B P is necessary but not sufficient for Q.
  - C P is sufficient but not necessary for Q.
  - D P is neither necessary nor sufficient for Q.
- 6 Which of the following statements is/are true?
- I For every real number  $x$ , there exists a real number  $y$  such that  $y > x$ .
  - II There exists a real number  $y$  such that for every real number  $x$ ,  $y > x$ .
  - III For every real number  $x$ , there exists a real number  $y$  such that  $xy = 1$ .
- A none of them
  - B I only
  - C II only
  - D III only
  - E I and II only
  - F I and III only
  - G II and III only
  - H I, II and III
- 7 A teacher says: '**You will pass only if you revise.**' Which of the following statements means the same thing?
- A If you pass, then you revised.
  - B If you revise, then you will pass.
  - C You will pass if you revise.
  - D If you do not pass, then you did not revise.
  - E You will pass if and only if you revise.

8 Consider the statement: **there exists a real number  $k$  such that for all real  $x$ ,  $x^2 + kx + 1 > 0$ .** Which of the following is true?

- A The statement is true; for example  $k = 0$  works.
- B The statement is false because  $x^2 + kx + 1$  can be negative for large  $x$ .
- C The statement is true; every value of  $k$  works.
- D The statement is false because for every  $k$  there is an  $x$  with  $x^2 + kx + 1 \leq 0$ .
- E The statement is true; for example  $k = 2$  works.

9 Which of the following statements is/are true about real numbers?

- I For all  $x$  and all  $y$ , if  $x < y$  then  $x^2 < y^2$ .
- II For all  $x$  there exists  $y$  such that  $x < y$  and  $x^2 < y^2$ .
- III There exist  $x$  and  $y$  with  $x < y$  and  $x^2 > y^2$ .

- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

10 A statement reads: **for every positive integer  $n$ , there is a prime  $p$  with  $n < p < 2n$ .** Which of the following is its negation?

- A For every positive integer  $n$ , there is a prime  $p$  with  $p \leq n$  or  $p \geq 2n$ .
- B There is a positive integer  $n$  such that no prime  $p$  satisfies  $n < p < 2n$ .
- C There is a positive integer  $n$  and a prime  $p$  with  $p \leq n$  or  $p \geq 2n$ .
- D There is a prime  $p$  such that for every positive integer  $n$ ,  $n < p < 2n$  fails.
- E For every positive integer  $n$ , no prime  $p$  satisfies  $n < p < 2n$ .

**11** Which of the following statements is/are logically equivalent to '**if it is raining, then the ground is wet**'?

- I** It is not raining, or the ground is wet.
- II** If the ground is not wet, then it is not raining.
- III** If the ground is wet, then it is raining.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

**12** Consider the statement  $S$ : **for all real  $x$ , if  $x > 0$  then  $x + \frac{1}{x} \geq 2$** . Which of the following statements is/are true?

- I**  $S$  is true.
- II** The converse of  $S$  (for all real  $x$ , if  $x + \frac{1}{x} \geq 2$  then  $x > 0$ ) is true.
- III** The contrapositive of  $S$  is true.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III